CS24307 — THEORY (3.0 CR) MODULE I NOTES BIT MESRA | NEP SCHEME

Module 1: Intelligent Agents & PEAS

Foundations of AI, Rationality, PEAS & Environment Dimensions

Module I: Intelligent Agents & PEAS — Topics Covered

1. Definitions of AI & Turing Test
2. Concept of Rationality vs. Omniscience
3. PEAS Framework (Performance, Environment, Actuators, Sensors)
4. Environment Taxonomies (7 Dimensions)
5. Agent Structures: Reflex, Goal, Utility & Learning Agents

1. PEAS Framework Formulation

Agent Type	Performance Measure (P)	Environment (E)	Actuators (A)	Sensors (S)
Automated Taxi Driver	Safety, speed, comfort, maximum profit, legal compliance	Roads, traffic, pedestrians, weather conditions	Steering wheel, accelerator, brake, horn, display	Cameras, LiDAR, radar, GPS, speedometer, engine sensors
Medical Diagnosis System	Patient health recovery, minimized cost, zero misdiagnosis	Patient, hospital database, laboratory staff	Screen displays, prescription generation, referral alerts	Keyboard input of symptoms, patient test results

2. Environment Dimensions Taxonomy

- **Fully Observable vs. Partially Observable:** Agent sensors give complete access to world state vs. noisy/occluded states.
- **Deterministic vs. Stochastic:** Next state is completely determined by current state and action vs. randomness.
- **Episodic vs. Sequential:** Current decision does not affect future episodes vs. current actions dictate future rewards.
- **Static vs. Dynamic:** Environment does not change while agent is deliberating vs. continuously evolving.
- **Discrete vs. Continuous:** Finite number of states/actions (Chess) vs. continuous state/time (Driving).